Task 2: Exploratory Data Analysis (EDA) Report

Project: Web Scraping and Exploratory Data Analysis
Dataset: Books Dataset
Data source: Books to Scrape (books.toscrape.com)
Number of records: 100 books
Tools used: Python, Google Colab, Pandas, Matplotlib, SciPy

1. Introduction

The purpose of this project was to perform Exploratory Data Analysis (EDA) on a dataset collected from a public book-catalogue website using Python web-scraping techniques. The analysis aimed to understand the dataset's structure, identify patterns in book prices and ratings, test a statistical hypothesis, and detect potential data-quality issues.

2. Dataset Description

The dataset contained 100 book records and six columns: Title, Price, Rating, Availability, Book_URL, and Price_numeric. The dataset was processed using Pandas, while Matplotlib was used to create visualizations.

The Price_numeric column was used for numerical calculations because the original Price column contained currency-formatted text. Book ratings were represented as five categories: One, Two, Three, Four, and Five.

3. Data Quality Assessment

The dataset was checked for missing values, duplicate records, duplicate titles, invalid prices, unexpected ratings, and empty titles or URLs.

The checks found no missing values in any of the six columns. There were zero duplicate rows and zero duplicate book titles. All prices were positive, all ratings matched the five expected categories, and no titles or URLs were empty. All 100 books were recorded as in stock.

These results indicate that the dataset passed the data-quality checks performed. However, these checks do not guarantee that every field is factually correct.

4. Descriptive Statistical Analysis

The descriptive statistics for book prices were:

Statistic	Value
Number of books	100
Mean price	£34.56

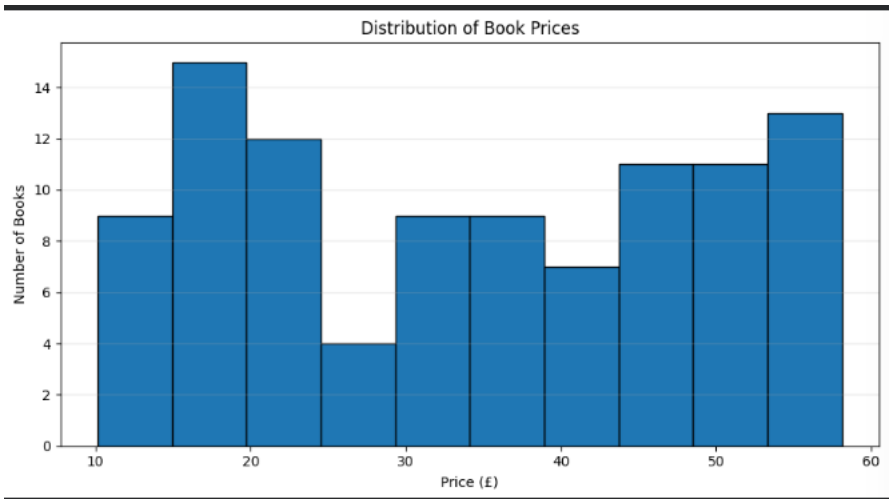
Median price	£34.78
Standard deviation	£14.64
Minimum price	£10.16
First quartile	£19.90
Third quartile	£47.97
Maximum price	£58.11

The mean and median prices were similar, at £34.56 and £34.78 respectively. The price range was £47.95, showing considerable variation between the least expensive and most expensive books.

5. Visualization and Pattern Identification

5.1 Distribution of Book Prices

A histogram was created to examine the distribution of prices. The chart showed prices ranging from approximately £10 to £58, with variation across the price intervals and visible concentrations in lower- and higher-price groups.



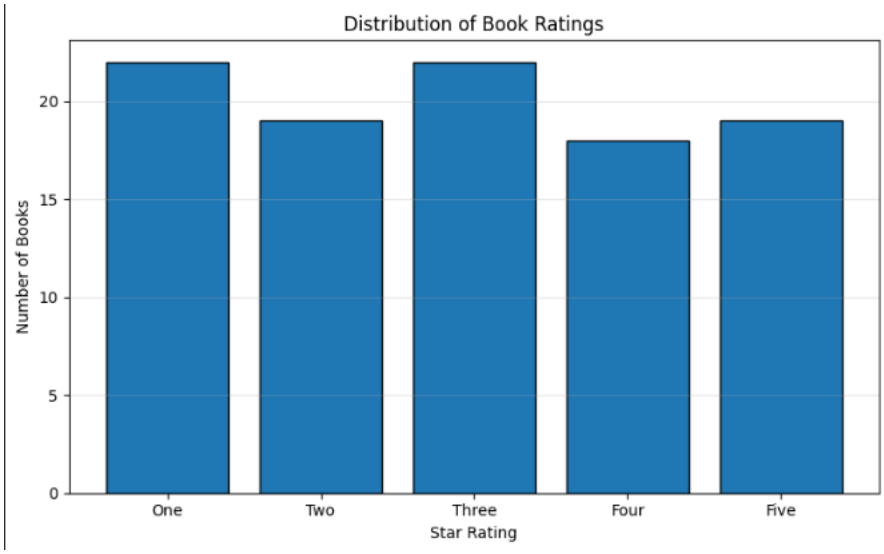
5.2 Distribution of Book Ratings

A bar chart was used to examine the frequency of each star rating.

Rating	Number of books	Percentage
One star	22	22%
Two stars	19	19%
Three stars	22	22%

Four stars	18	18%
Five stars	19	19%

One-star and three-star books were the most frequent categories, each representing 22% of the dataset. Four-star books were the least frequent, representing 18%. Overall, the rating distribution was relatively balanced.



5.3 Book Prices by Star Rating

A box plot was created to compare price distributions across rating categories. The five-star group appeared to have a lower median price than the other groups. However, visual differences alone cannot establish that prices differ statistically between ratings.



6. Hypothesis Testing

The research question was: **Do book prices differ across the five star-rating groups?**

The hypotheses were:

- **Null hypothesis (H_0):** Book-price distributions are the same across the five rating groups.
- **Alternative hypothesis (H_1):** At least one rating group has a different book-price distribution.

A Kruskal–Wallis test was conducted at a significance level of 0.05.

Test result	**Value**
H-statistic	2.6655
P-value	0.6153
Significance level	0.05

Since the p-value of 0.6153 was greater than 0.05, the null hypothesis was not rejected. The analysis did not find statistically significant evidence of differences in book-price distributions across the five rating groups.

This result does not prove that all rating groups have identical price distributions.

7. Outlier Detection

The Interquartile Range (IQR) method was used to identify potential price outliers.

The first quartile was £19.90, the third quartile was £47.97, and the IQR was £28.07. The calculated lower boundary was −£22.21, while the upper boundary was £90.07.

No book prices fell outside these boundaries. Therefore, no potential price outliers were detected using the IQR method.

8. Limitations

The dataset contained 100 books collected from the website and may not represent every book available online. The analysis describes the collected sample rather than the entire book market.

The availability field showed all books as in stock, but availability may change over time. In addition, the hypothesis test did not establish that the rating groups have identical price distributions. Further analysis with additional data could provide more insight into the relationship between prices and ratings.

9. Conclusion

The exploratory analysis examined the structure, quality, price distribution, and ratings of 100 books collected through web scraping.

The dataset contained no missing values, duplicate rows, duplicate titles, invalid prices, unexpected ratings, or empty titles and URLs in the checks performed. The average book price was £34.56, while the median was £34.78. One-star and three-star books were the most common rating categories, each accounting for 22% of the dataset.

The IQR method detected no potential price outliers. The Kruskal–Wallis test found no statistically significant evidence of differences in price distributions across the rating groups at the 5% significance level.

Overall, the analysis demonstrated how Python can be used to inspect a dataset, summarize numerical variables, visualize patterns, test a hypothesis, and investigate potential data-quality problems.

10. Tools and Libraries

- **Python:** Main programming language.
- **Requests:** Retrieved website HTML.
- **BeautifulSoup:** Extracted book information from HTML.
- **Pandas:** Organized and analyzed the dataset.
- **Matplotlib:** Created charts and visualizations.
- **SciPy:** Performed the Kruskal–Wallis statistical test.
- **Google Colab:** Provided the notebook environment for running the code.